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Optimal use of dual orexin receptor antagonists for insomnia: a network meta-analysis perspective

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Dear Editor

Recently, we published a systematic review and a random-effects model network meta-analysis in *Translational Psychiatry*, which included double-blind, randomized, placebo-controlled trials (DBRPCTs) of dual orexin receptor antagonists (DORAs), namely lemborexant (LEM), suvorexant (SUV), and daridorexant (DAR) [1]. This previous network meta-analysis revealed significant differences in the efficacy of DORAs for sleep onset and sleep maintenance [1]. This year, two DBRPCTs of LEM or DAR were reported in China (Table S1) [2, 3]. Meta-analysis enhances statistical power by aggregating data and allows detection of differences that may be obscured in individual trials; thus, we conducted an updated network meta-analysis to strengthen the robustness of previously reported findings [4]. Moreover, this updated network meta-analysis included adverse event outcomes, including abnormal dreams/nightmares, sleep paralysis, and withdrawal symptoms, allowing for a more comprehensive assessment of both efficacy and safety.

This study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (Table S2) and Cochrane Handbook for Systematic Reviews of Interventions [4] and was registered on the Open Science Framework (<https://osf.io/r6y5w>). At least two of the authors verified the literature search, data transfer accuracy, and calculations. Detailed information regarding the search strategy is shown in Figure S1.

The efficacy outcomes of this systematic review and meta-analysis were subjective time to sleep onset (sTSO) at month 1 (primary), subjective total sleep time (sTST) at month 1 (co-primary), subjective wake after sleep onset (sWASO) at month 1, and Insomnia Severity Index (ISI) [5] scores at month 1. Other outcomes included all-cause discontinuation, adverse event-related discontinuation, and the incidence of at least one adverse event, somnolence, dizziness, falls, sleep paralysis, abnormal dreams/nightmares, headache, nasopharyngitis, upper respiratory tract infection, and withdrawal symptom questionnaire [6] scores during the run-out phase. The algebraic sign of the numerical scores for sTST at month 1 was reversed because lower scores indicated a greater impairment. We conducted a network meta-analysis for the outcomes that included at least five trials. The intention-to-treat or modified intention-to-treat principles were employed to analyze extracted data. The complete analysis data was not used. Missing required data from the studies were searched in published systematic review articles [1].

A frequentist network meta-analysis with a random-effects model was conducted to pool standardized mean differences (SMDs) and odds ratios (ORs) with corresponding 95% confidence intervals (CIs) [7]. The network heterogeneity was assessed with τ^2 statistics. The design-by-treatment test (globally) and the Separating Indirect from

Direct Evidence test (locally) were performed to statistically evaluate incoherence. Surface under the curve cumulative ranking (SUCRA) probabilities were used to rank treatments for each outcome. We identified the sufficiency of the distribution differences to validate the analysis by comparing the distribution of possible effect modifiers across included treatments in the network meta-analysis using the Kruskal–Wallis test (continuous variables) and the Pearson chi-square test or Fisher's exact test (categorical variables) and by assessing their actual influence on the treatment effect through network meta-regression analyses. The proportion of females, mean age, proportion of older people (aged ≥ 65 years), total number of participants, and publication year were potential confounding factors (Table S3). We conducted a network meta-analysis for acceptability, tolerability, and safety outcomes—where observation periods could not be matched—excluding one long-duration (6 months) study of LEM [8] similar to the recent network meta-analysis that evaluated any outcomes divided into acute treatment (< 3 months) or long-term treatment (≥ 3 months)(i.e., our network meta-analysis included only 3 month or shorter studies) [9]. We assessed the overall risk of bias for each trial in our systematic review using the Cochrane risk of bias tool for randomized trials version 2 (<https://www.riskofbias.info/>). Finally, the results were integrated into the Confidence in Network Meta-Analysis application, which is an adaptation of the Grading of Recommendations Assessment, Development, and Evaluation approach, to examine the credibility of each network meta-analysis [10]. Publication bias was planned to be assessed using a funnel plot.

Figure S1 shows a flowchart of the literature search along with a detailed explanation of the process. Finally, the meta-analysis included 10 DBRPCTs ($n = 5597$; average age = 55.46 years; 68.03% female) [2, 3, 8, 11–15]. The treatment arms included DAR 25 mg/day (DAR25), DAR 50 mg/day (DAR50), LEM 5 mg/day (LEM5), LEM 10 mg/day (LEM10), SUV 20 mg/day (15 mg/day for people aged ≥ 65 years, SUV20/15), and placebo. Table S1 summarizes the characteristics of the included studies. The overall risk of bias in all studies was rated as “Low” (Table S4). No evidence of transitivity violations was observed when examining study characteristics across different trials (Table S3).

All active treatments demonstrated superior efficacy compared to placebo across outcomes (Figure S2 and Supplementary Appendix 1–4). The ranges of SMDs (95% CIs) at month 1 were as follows: sTSO at month 1, -0.399 ($-0.527, -0.272$) for LEM10 to -0.164 ($-0.299, -0.028$) for SUV20/15; sTST at month 1, -0.441 ($-0.549, -0.332$) for DAR50 to -0.196 ($-0.317, -0.075$) for LEM5; sWASO at month 1, -0.223 ($-0.336, -0.110$) for LEM10 to -0.121 ($-0.240, -0.002$) for LEM5; and ISI score at month 1; -0.345 ($-0.459, -0.232$) for LEM10 to -0.222 ($-0.317, -0.128$) for DAR25.

LEM10 was associated with improved sTSO at month 1 compared with those of DAR25 and SUV20/15 (Supplementary Appendix 1). DAR50 was associated with improved sTST at month 1 compared with those of DAR25 and

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LEM5 (Supplementary Appendix 2). LEM10 improved sTST at month 1 compared with those of LEM5 (Supplementary Appendix 2). The other two efficacy outcomes demonstrated no significant difference among the active treatments (Supplementary Appendix 3–4).

LEM10 exhibited the highest SUCRA values across all efficacy outcomes, with the exception of sTST at Month 1, where DAR50 showed the highest value. Network meta-regression analyses identified no potential confounding factors affecting the effect sizes of sTSO and sTST at month 1 (Supplementary Appendix 1–2).

DRA25, DAR50, LEM10, and SUV20/15 were associated with a higher incidence of somnolence than placebo (Supplementary Appendix 8). However, no significant differences in acceptability, tolerability, or any other safety outcomes were reported between any active treatments and the placebo (Supplementary Appendix 5–7, 9–16).

We did not perform a network meta-analysis for death, suicidal ideation and/or behavior, or cataplexy because of their rarity (Supplementary Appendix 17).

Supplementary Appendices S1–S16 present the results for heterogeneity and inconsistency for all outcomes. Overall confidence in the evidence was generally rated as low or very low. The limitations of our study were shown in Supplementary Text.

The present meta-analysis incorporated two studies in addition to those in the previous analysis, resulting in a total of 10 DBRPCTs comprising 5597 patients. The previous network meta-analysis reported inconsistency and/or heterogeneity across multiple outcomes [1]; however, the present network meta-analysis identified such issues in fewer outcomes, likely indicating the increased sample size and improved statistical power. Despite this, the findings of efficacy, acceptability, and tolerability outcomes were consistent with those of the previous network meta-analysis [1].

This current network meta-analysis assessed sleep paralysis, abnormal dreams/nightmares, and withdrawal symptoms during the run-out phase, which could not be evaluated in the previous network meta-analysis due to insufficient data [1]. The present network meta-analysis revealed no significant differences in the incidence of sleep paralysis and abnormal dreams/nightmares between any active treatment and placebo. Nevertheless, the incidence of sleep paralysis and abnormal dreams or nightmares associated with these DORAs was rare (Supplementary Appendix 17). In addition, DAR25, DAR50, LEM5, and LEM10 were not associated with withdrawal symptoms during the run-out phase, although the SUV studies included in our network meta-analysis did not provide data on this outcome.

Comparison of LEM10 with placebo revealed local inconsistency for sTSO at month 1. Therefore, clinicians and patients should rely on the SMD from the direct comparison (-0.403 ; 95% CI: -0.531 to -0.276) rather than the estimate derived from the network meta-analysis (Supplementary Appendix 1).

DATA SHARING

Data used for the current study were reported in articles as cited in this paper.

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AUTHOR CONTRIBUTIONS

TK had full access to all the data in the study and takes responsibility for the integrity of the data along with the accuracy of the data analysis. TK was involved in the study concept and design. The literature search, data transfer accuracy, and acquisition and calculations of data were conducted by at least two of the authors (TK, MH, and KMS). TK and TI performed the statistical analysis. All the authors performed interpretation of the data, and wrote the manuscript. NI supervised the review.

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COMPETING INTEREST

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ADDITIONAL INFORMATION

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